

ARCHITECTURE · ENTERPRISE IT · AGENTIC ERA

AIDEX

*An Architecture for Human-Curated,
AI-Enabled Knowledge Work*

James D. Longmire

Northrop Grumman Fellow · Chief Architect, Digital Ecosystems

Independent research. Does not represent any employer or program.

01 The Strategic Claim

The defense industrial base is in a workforce transition, and the substrate of knowledge work is changing at the same time.

01

Workforce transition is real

Senior engineers retire faster than they can be replaced. Mid-career talent moves. Early-career hires inherit institutional context that took fifteen years to absorb.

02

The substrate has changed

Agentic capabilities, synthesis, and collaboration substrates compress what was assembly, research, and coordination. The shape of knowledge work is different now.

03

Accountability is the frontier

Consumer-grade alternatives are not building for accountability. Enterprises that deploy HCAE-aligned substrate deliberately position at a frontier competitors have not reached.

The decision is whether to face this transition with a framework, a subdomain, and a control plane, or without them.

02 Three Terms, One Architecture

FRAMEWORK

HCAE

Human-Curated, AI-Enabled

The practice discipline. The human practitioner curates AI-enabled output, remains the locus of judgment and accountability, and contributes to institutional practice as an attributed and consenting participant.

SUBDOMAIN

AIDEX

AI Digital Experience

The worker-facing substrate. The architecture that expresses HCAE operationally at the digital experience layer. One of nine peer subdomains under AEON.

CONTROL PLANE

AEON

AI Enterprise Orchestration Nexus

The enterprise control plane for the agentic era. Identity, authority, policy, evidence, integration, and capability composition across all subdomains.

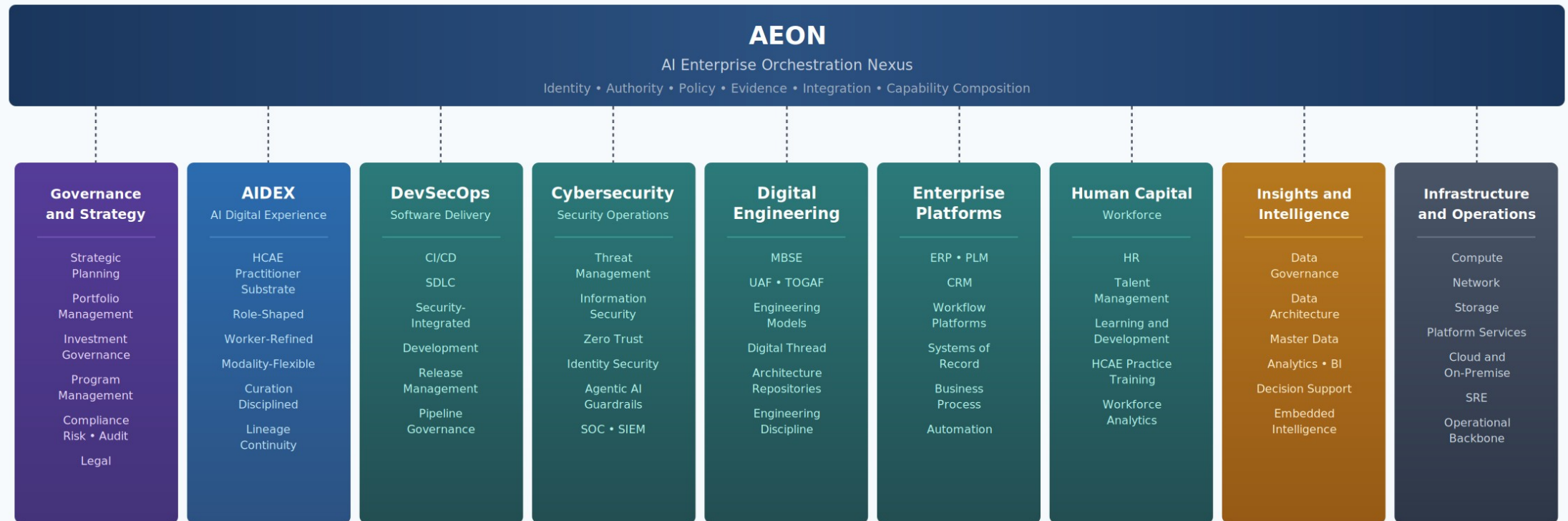
The next-generation knowledge worker is the HCAE practitioner operating on the AIDEX substrate under AEON orchestration.

03 Where AIDEX Sits

AEON orchestrates nine peer subdomains. AIDEX is the worker-facing AI digital experience.

AEON Enterprise Architecture Viewpoint

AI Enterprise Orchestration Nexus over Peer Subdomain Capabilities



Subdomains operate as peers under AEON orchestration. Identity, policy, evidence, and integration fabric flow across all subdomains. HCAE is the practice framework AIDEX expresses; HCAE-aligned discipline extends to worker engagement with all subdomains.

04 HCAE: The Practice Framework

Human- Curated, AI-Enabled.

The order is not cosmetic.

Human-curated comes first because curation anchors accountability. AI-enabled comes second because enablement is the function of the substrate.

Curation, not approval

The practitioner engages continuously with substrate output throughout the work. She selects, attributes, revises, and signs. The signature certifies her curation.

Risk-tiered curation

Per-action confirmation for consequential work. Batch confirmation for routine work. Sampled audit for low-stakes work. The tier in effect is captured in the audit trail.

Accountability is architectural

Not a policy commitment. Every consequential action traces to a human principal through a cryptographically verifiable chain. The substrate enforces it; AEON provides the fabric.

05 AIDEX: The Worker-Facing Substrate

What AIDEX does for the practitioner, expressed as the propose-and-confirm posture under HCAE discipline.

WHAT AIDEX PROVIDES

- **Role-shaped substrate**
Configured to the worker's actual role, not a generic chat surface
- **Modality-flexible interaction**
Chat, voice, document, agentic execution across the same session
- **Propose-and-confirm posture**
Consequential actions surface as proposals; curation is captured at the moment
- **Worker-owned memory**
Personalization to the practitioner over time, under her control and consent
- **Lineage continuity**
Attributed contribution to her role's accumulated practice across tenure boundaries
- **Cross-modality state**
Context carries between surfaces without forcing the worker to reassemble it

WHAT HCAE PRESERVES

- **Judgment**
The practitioner remains the locus of professional judgment, not the substrate
- **Accountability**
Actions trace to a human principal who authorized them, not to a dissolving system
- **Professional standing**
The worker's signature certifies her curation, not the substrate's production
- **Institutional discipline**
Curation tier is recorded; the institution can verify what discipline applied
- **Attribution**
Contributions to lineage carry the contributor's identity as an attestation
- **Consent**
Memory and lineage are opt-in, worker-controlled, and revocable within retention limits

06 The Modularity Model

Eight independently varying axes. Reconfiguration along one axis does not disturb the others, a precondition for maintainability over defense program timescales.

1

Presentation

Voice, self-reference, visual identity

2

Persona

Behavioral character, register, posture

3

Role

Role-specific knowledge, tools, workflows

4

Authority

Policy that AEON evaluates per action

5

Context

Working state across modality surfaces

6

Memory

Worker-owned, opt-in personalization

7

Modality

Chat, voice, document, agentic execution

8

Lineage

Attributed contribution across tenure

07 AEON: The Enterprise Control Plane

AIDEX consumes AEON services. AEON specifies them as enterprise-wide planes.

Identity

IDAM federation, CAC/PIV, cross-classification continuity

Authority

Per-action policy decisions on identity-resolved attributes

Policy

Decomposition from enterprise scope into subdomain enforcement

Evidence

Aggregated audit picture across all subdomains, cryptographically chained

Integration

Cross-subdomain workflow composition under principal delegation

Capability

Lifecycle, composition, and governance of agentic capabilities

AIDEX does not maintain **its own user store, its own policy engine, its own audit infrastructure, or its own vendor governance**. AEON provides these as services that AIDEX consumes.

08 Cowork as Deployed Reference Architecture

Claude Cowork (Anthropic, GA April 2026) is the working reference for the productivity layer. AIDEX is not Cowork, but the concept can be developed against a deployed baseline rather than against speculation.

What Cowork Demonstrates	Where Cowork Falls Short on HCAE	The Migration Path
● Sandboxed VM, MCP integration, plugin architecture ● Enterprise governance posture and audit hooks ● 1M context, multi-modal interaction ● Outcome-oriented agentic execution at scale	● No deep IDAM binding to enterprise identity ● No long-arc worker-owned memory layer ● No attributed lineage continuity ● Partial classification deployment story	● Phase 1: deploy Cowork-class baseline ● Phase 2: AEON control plane integration ● Phase 3: role packaging and SoR integration ● Phase 4-6: memory, lineage, multi-backend

Substrate portability is an architectural commitment. The next generation will be different; the architecture has to survive the transition.

09 Multi-Backend Deployment Topology

AIDEX supports multiple model backends simultaneously, mapped to classification environments. Not a binary cloud-vs-on-premises choice.

UNCLASSIFIED

Commercial Cloud

Frontier commercial models. Highest capability. Strategic and administrative work outside controlled environments.

CUI / FEDRAMP HIGH

GovCloud Commercial

Same vendor models in compliant clouds. Capability-comparable to commercial. Controlled unclassified work under federal compliance.

NOFORN / AIR-GAPPED

On-Prem Open-Weight

Llama-class open-weight in customer-controlled environments. Capability gap real but narrowing. Western defense default.

ENCLAVED / CLASSIFIED

On-Prem Commercial

Vendor-licensed commercial models deployed in enclaves. Where available; vendor and program-specific arrangement.

Workers experience identity continuity across environments. AEON's identity plane federates; AIDEX presents a coherent substrate.

10 The Accountability Frontier

*Consumer-grade alternatives are not building for accountability.
The consumer market does not require it.*

The defense enterprise that deploys substrate without HCAE-aligned accountability is deploying a system that may produce outputs the institution cannot govern.

Workforce continuity

Lineage compounds across tenure; institutional knowledge stops evaporating at retirement

Decision quality

Curation discipline raises the floor on what reaches signed work

Positional advantage

Built infrastructure competitors have not built and the consumer market is not building

11 What This Asks of the Enterprise

Enterprise IT leadership is the natural owner. Success depends on partnership across the institution.

OWNER	WORKER OWNERSHIP	PRACTICE	DEPLOYMENT
Enterprise IT	Human Resources	Chief Learning Officer	Program Leadership
WHAT'S ASKED	WHAT'S ASKED	WHAT'S ASKED	WHAT'S ASKED
— AEON control plane integration	— Memory and lineage consent framework	— HCAE practice training program	— Pilot role and program selection
— Multi-backend deployment topology	— Contribution attribution policy	— Curation discipline as professional skill	— Systems of record integration scope
— AIDEX substrate package and lifecycle	— Worker contribution revocation rights	— Role-based practitioner certification	— Operational metrics and validation

12 Decision Frame

The next step is a deployment design.

- 01 Select a target role and a target program
- 02 Configure the backend topology for that program's classification environment
- 03 Integrate AIDEX with AEON's control plane
- 04 Establish HCAE training and worker-ownership commitments in policy and contract
- 05 Instrument metrics that validate (or falsify) the strategic argument
- 06 Test the modularity model against actual practice

*The workforce transition is happening on its own timeline.
The question is whether to face it with the framework, the subdomain, and the control plane, or without them.*